

VETO: Evidence-Constrained Harness Selection for Reliable Chart Reasoning

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Abstract

*Optimizing an executable harness for current task accuracy does not explicitly test whether it responds correctly when answer-relevant visual evidence changes. We study paired visual evidence as a selection criterion before a subsequent weight update. VETO retains the incoming harness and selects among candidates whose empirical paired correctness does not decrease. The weight update reuses WHALE’s native rejection-sampling fine-tuning. Our prospective study uses a shared scoring protocol, matched candidate archives and independently seeded training branches. **Study status: experiments are incomplete; this draft does not claim a measured improvement.***

1. Introduction

Chart reasoning requires answers to follow the information in the presented chart. Counterfactual chart evaluation already exposes failures hidden by single-chart scores [3]. Harness optimization and non-regression acceptance also have established precedents [1, 2]. We examine the narrower question of whether paired evidence is useful for choosing the harness that supplies subsequent successful training trajectories.

Our intended contribution is an experimentally supported selection mechanism, not the invention of harness-weight alternation or counterfactual evaluation. We use one complete training–search–training cycle and compare the paired gate against a same-data single-image accuracy gate and counterfactual augmentation. No long-horizon causal propagation or population reliability guarantee is assumed.

2. Method

Let θ denote fixed model weights, h_{in} the phase incoming harness, and $C = \{(x_i, x'_i, q_i, y_i, y'_i)\}$ valid answer-changing image pairs. With correctness indicators c_i and c'_i , define

$$P_C(\theta, h) = |C|^{-1} \sum_i c_i c'_i. \quad (1)$$

The eligible candidate set is

$$\mathcal{H}_{\text{ok}} = \{h : P_C(\theta, h) \geq P_C(\theta, h_{\text{in}}) - \epsilon\} \cup \{h_{\text{in}}\}. \quad (2)$$

We fix $\epsilon = 0$ and select lexicographically:

$$h_{\text{next}} = \arg \max_{h \in \mathcal{H}_{\text{ok}}} (A_H(\theta, h), -\text{turns}_H(\theta, h)). \quad (3)$$

Exact ties retain the original archive order. A subsequent native RSFT stage uses successful trajectories under h_{next} ; its objective and success filter are unchanged. These empirical constraints do not guarantee independent test accuracy or non-regression after training.

The answer parser and verifier are shared across all conditions and cannot be changed by candidates. Candidate code runs with file and network confinement. Current weights, incoming harness, audit sample identities and decoding settings are fixed within a selection phase.

To separate the gate from an ordinary accuracy check, let M_C be single-image accuracy on the same paired data and q_0 the fraction of pairs with both answers incorrect. Expanding $(1 - c_i)(1 - c'_i)$ gives the exact identity

$$P_C = 2M_C - 1 + q_0, \quad \Delta P_C = 2\Delta M_C + \Delta q_0.$$

Thus equal single-image accuracy need not imply equal paired correctness. If q_0 is constant across the fixed-weight candidate archive, the two zero-tolerance gates are equivalent on that archive. Conversely, variation in q_0 alone does not establish a different selected harness or a downstream benefit. At fixed M_C , larger P_C also entails more both-incorrect pairs; the joint endpoint must therefore be interpreted alongside ordinary accuracy, rather than as a guarantee that every type of error decreases. Joint correctness does not identify why an answer fails. Perception, arithmetic, instruction following and answer formatting can all affect this endpoint; paired failures alone do not establish visual non-use or a latent mechanism.

3. Experimental Protocol

We use the public post-trained Qwen3.5-4B checkpoint [6] and seeds 42, 43, 44. Each weight stage has four batches,

VETO: test visual evidence before choosing the next training harness

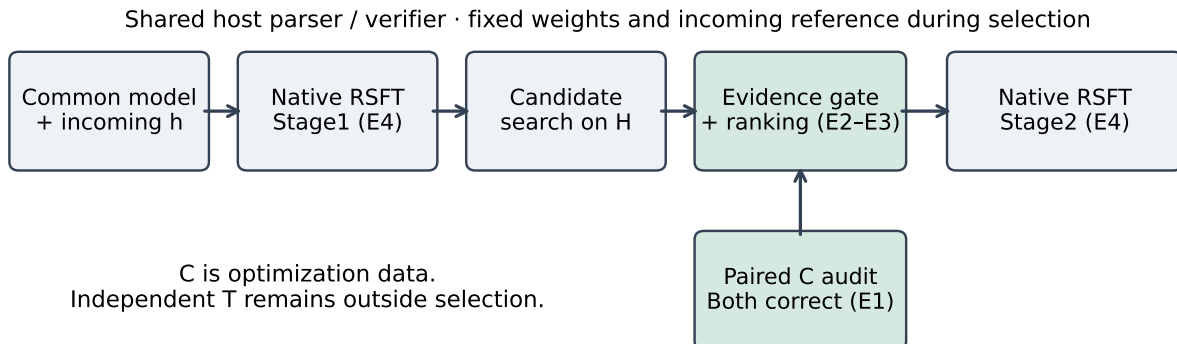


Figure 1. Paired evidence constrains the harness selected for the next native RSFT stage. The shared scorer, candidate search and weight optimizer are common infrastructure.

Condition	Single image	Both images
Weight only	—	—
Harness only	—	—
WHALE, one cycle	—	—
VETO, one cycle	—	—

Table 1. Template only. No formal condition has a completed scientific result. Final tables will include all seeds, mean, sample standard deviation and source-cluster confidence intervals.

Selection / control	Paired score	Difference
Ordinary accuracy	—	—
Same-C marginal gate	—	—
VETO paired gate	—	—
50% paired augmentation	—	—

Table 2. Mandatory controls, pending real continuation experiments. Equal decisions will be identified explicitly.

Condition	Human	Augmented	Cost
WHALE	—	—	—
VETO	—	—	—
Controls	—	—	—

Table 3. ChartQA fixed subset and total GPU/API costs. No external model score is available in this scaffold.

070 eight questions per batch and eight trajectories per question.
 071 Each search attempts at most three candidates. Identical
 072 first stages and candidate measurements are shared within
 073 a seed; subsequent branches are evaluated as dependent,
 074 matched comparisons rather than independent first-stage tri-
 075 als.

076 PlotQA-EvidencePairs is a separately named dataset de-
 077 rived from PlotQA source tables [5], with comparison,
 078 value-reading and difference questions. All variants of a
 079 source table remain in the same partition: W2048, H128,
 080 C64 pairs, V256 pairs, T1024 pairs, and an independent
 081 recheck pool. C is optimization data. Numeric answers
 082 use a declared tolerance; answer-changing intervals must
 083 not overlap. The external ChartQA [4] evaluation uses a
 084 preregistered subset of 256 human and 256 augmented ques-
 085 tions; it is not the full official benchmark score.

086 The common initial prompt and learning rate are selected
 087 using ordinary development accuracy only. The final check-
 088 point is determined by the frozen budget. Test data are ex-
 089 cluded from all model and harness selection.

4. Results

The same-data marginal gate and counterfactual augmenta-
 tion are mandatory controls. If gates select the same har-
 ness, that observation provides no evidence for a distinct
 benefit of VETO. We check the error decomposition on the
 actual fixed-weight C archives; calibration measurements at
 different weights cannot establish that the acceptance rules
 choose different harnesses.

Stage trajectories and performance–cost plots will be
 generated from complete branch records. Empty plots or
 simulated improvements are not experimental evidence.

5. Limitations and Decision Criteria

The study is limited to one model size, one domain and one complete alternation. Source-derived renderings do not establish general visual-agent reliability. The native checkpoint protocol omits Adam moments; all branches use the same restoration behavior. We report full allocation costs including failed runs, loading, candidate evaluation and audit overhead.

The prospective candidate criterion requires at least one percentage point mean paired improvement, nonnegative differences across all three seeds, positive differences in at least two seeds, ordinary accuracy loss no greater than one point, and a positive lower bound of the paired source-cluster95% interval. Independent value beyond a marginal gate must also be demonstrated. Unsupported outcomes will be reported as inconclusive rather than hidden.

References

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